

CLASSIFICATION

PERFORMANCE EVALUATION – PART II

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PERFORMANCE EVALUATION

The following concepts will be introduced:

✓ **PERFORMANCE OF A CLASSIFICATION MODEL**

- ACCURACY
- SPEED
- ROBUSTNESS
- SCALABILITY
- INTERPRETABILITY

✓ **HOLDOUT - ITERATED HOLDOUT**

✓ **CROSS VALIDATION**

- K FOLDS
- K FOLDS STRATIFIED
- LEAVE ONE OUT

A classification analysis consists of developing different classification models in terms of both the type of the classification model and the associated parameters value. Such a procedure aims to “**SELECT**” a specific instance of a classification model, algorithm and parameters value, capable to “**ENSURE**” the “**MAXIMUM POSSIBLE PREDICTIVE ACCURACY**”.

Classification models are compared in terms of:

- ✓ **ACCURACY**
- ✓ **SPEED**
- ✓ **ROBUSTNESS**
- ✓ **SCALABILITY**
- ✓ **INTERPRETABILITY**

It is important because it

- measures the capability of the classification model to give reliable predictions on new records (i.e. records not available when the model was developed),
- allows to “**SELECT**” the instance of the classification model which likely provide the “**BEST PREDICTION PERFORMANCE**” on new records.

We use the following notation

D_T training set, consists of t records

D_{TS} test set, consists of v records

$$D = D_T \cup D_{TS}, D_T \cap D_{TS} = \emptyset, m = t + v$$

A good indicator of the accuracy achieved by a classifier is represented by the percentage of test record D_{TS} which are correctly classified.

We let y_i be the class value associated with the instance $\underline{x}_i \in D_{TS}$ and $f(\underline{x}_i)$ be the class value for the same instance as predicted by the classification model, then we can use the following loss function

$$L(y_i, f(\underline{x}_i)) = \begin{cases} 0 & \text{se } y_i = f(\underline{x}_i) \\ 1 & \text{se } y_i \neq f(\underline{x}_i) \end{cases}$$

to compute the **ACCURACY** as follows

$$\text{acc}(D_{TS}) = 1 - \frac{1}{V} \sum_{i=1}^V L(y_i, f(\underline{x}_i))$$

In some case we prefer to use the **ERROR**

$$\text{err}(D_{TS}) = 1 - \text{acc}(D_{TS}) = \frac{1}{V} \sum_{i=1}^V L(y_i, f(\underline{x}_i))$$

Classification algorithms differ from:

- ✓ **LEARNING TIME**

- ✓ **SPACE MEMORY**

Selecting the type of classification model to learn depends on the above aspects.

A classifier which requires high learning time and/or great memory space to be learnt can be learnt after **SAMPLING THE ORIGINAL DATASET**.

In such a case we accept not to exploit all the available information in spite of learning a given type of classification model that we think will ensure good performance measures.

ROBUSTNESS, SCALABILITY AND INTERPRETABILITY

A classification model/algorithm can be **ROBUST** or not with respect to

- ✓ **OUTLIERS**
- ✓ **MISSING DATA**
- ✓ **VARIATION OF TRAINING AND TEST SET DATA**

We say that a classifier is **SCALABLE** if it is capable to learn from huge amount of data. This property is intrinsically connected to the learning speed.

When Machine Learning is oriented towards **INTERPRETABILITY** of the problem to be solved and it is not limited to ensure good accuracy values, it is relevant the extracted rules to be simple and clearly understood by the **DOMAIN EXPERT** for the problem under study.

Account Length	VMail Message	Day Mins	Churn	Intl Calls	Intl Charge	State	Area Code	Phone
128	25	265.1	n	3	2.7	KS	415	382-4657
107	26	161.6	n	3	3.7	OH	415	371-7191
137	0	243.4	n	5	3.29	NJ	415	358-1921
84	0	299.4	y	7	1.78	OH	408	375-9999
75	0	166.7	n	3	2.73	OK	415	330-6626
118	0	223.4	n	6	1.7	KS	510	391-8027
121	24	218.2	n	7	2.03	MA	510	355-9993
147	0	157	n	6	1.92	MO	415	329-9001
117	0	184.5	n	4	2.35	KS	408	335-4719
141	37	258.6	n	5	3.02	WV	415	330-8173



You know that a **POINT ESTIMATE OF THE CLASSIFICATION MODEL'S ACCURACY IS NOT ENOUGH** to be confident that the Classification Model you developed will provide reliable predictions when queried on unseen records (new customers).

You also want to reduce the risk of **OVERFITTING/UNDERFITTING THE CUSTOMERS CALLS DATABASE**, provided by the **CEO OF THE TELECOM COMPANY**.

The dataset **D** is partitioned into disjoint subsets

D_T training set, consists of t records

D_{TS} test set, consists of v records

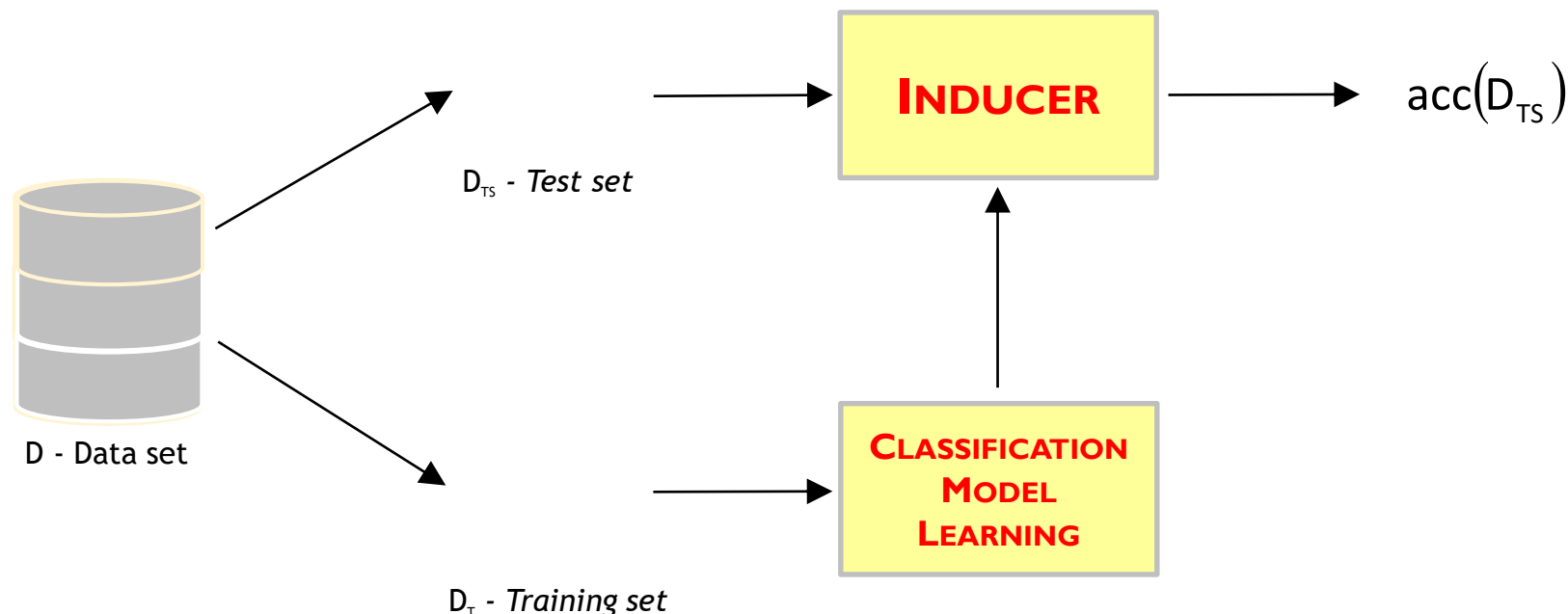
$$D = D_T \cup D_{TS}, D_T \cap D_{TS} = \emptyset, m = t + v$$

The **ACCURACY OF A CLASSIFICATION MODEL** is estimated by

$$\text{acc}(D_{TS}) = 1 - \frac{1}{v} \sum_{i=1}^v L(y_i, f(\underline{x}_i))$$

HOLDOUT partitions the dataset **D** in two subsets through a simple random sampling procedure. **Best practice** suggests **2/3 FOR THE TRAINING SET** and **1/3 FOR THE TEST SET**.

HOLDOUT consists of limiting the data which are used to learn the classifier, i.e. some records are saved (hold back) to estimate the reliability level of the classifier for the task to be solved.



The accuracy estimate depends on the choice of the test set, thus we can over(under)estimate the true value of the classification accuracy. A more robust estimate can be obtained through **ITERATED HOLDOUT** and **CROSS VALIDATION**.

Consists of iterating R times the Holdout method.

For each iteration r we extract a random sample D_{T_r} consisting of t records. We obtain

$$D_{TS_r} = D - D_{T_r}$$

The procedure is repeated R times and the accuracy of the classifier is estimated by the sample mean of the accuracy values $\text{acc}(D_{TS_r})$ computed on each test set D_{TS_r} :

$$\text{acc} = \frac{1}{R} \sum_{r=1}^R \text{acc}(D_{TS_r})$$

The number of iterations R can be selected exploiting specific statistical techniques.

ITERATED HOLDOUT SIGNIFICANTLY IMPROVES ON HOLDOUT, it estimates performance measures with a bias which is smaller than the bias associated with estimates obtained from the Holdout.

However, **ITERATED HOLDOUT** does not allow to control the number of times a given record is contained in the training and in the test set.

This could result in a strong bias with specific reference to cases where the data set contains dominant records (i.e. outliers).

In such a cases we resort to use more effective estimation schema, i.e. capable to control (reduce) the impact of outlier record(s) on the estimated value for the classifier's performance measures.

Ensures that each record of the dataset **D** is included into the training sets the same number of times and exactly one time in the test set.

The dataset **D** is partitioned into **K** disjoint subsets, exhaustive (a partition of **D**) and with almost a constant number of records

$$D_1, D_2, \dots, D_K$$

We perform **K** learning-testing iterations. At the **k-th** iteration we have

$$D_{T_k} = \{D_1, \dots, D_{k-1}, D_{k+1}, \dots, D_K\} \quad D_{TS_k} = D_k$$

The set **D_{T_k}** is used for training while its complement set **D_k** is used for testing at the **k-th** iteration.

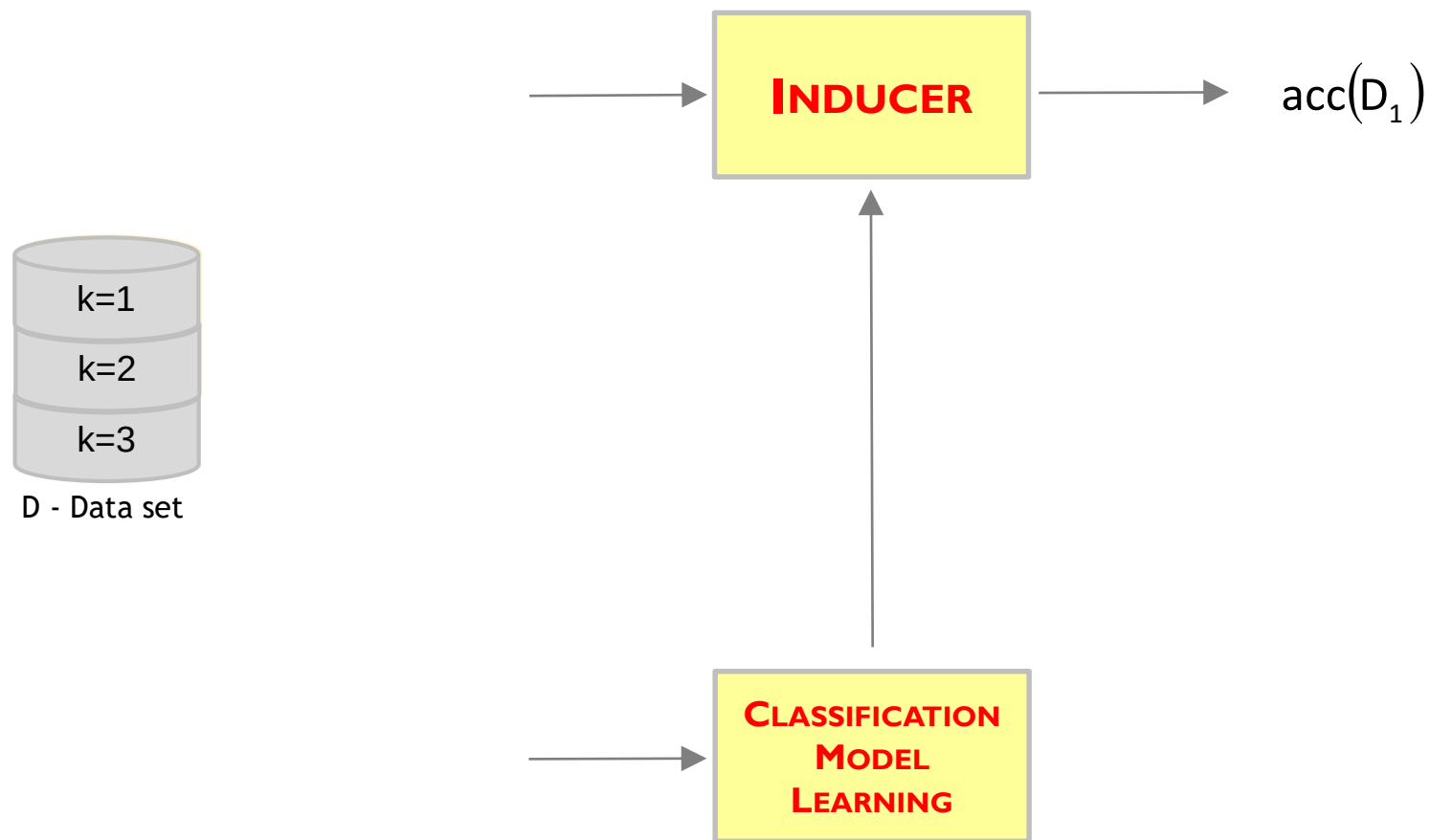
The classification algorithm undergoes **K** training-testing phases, while the **K** estimates of the classifier's accuracy used to compute the arithmetic mean

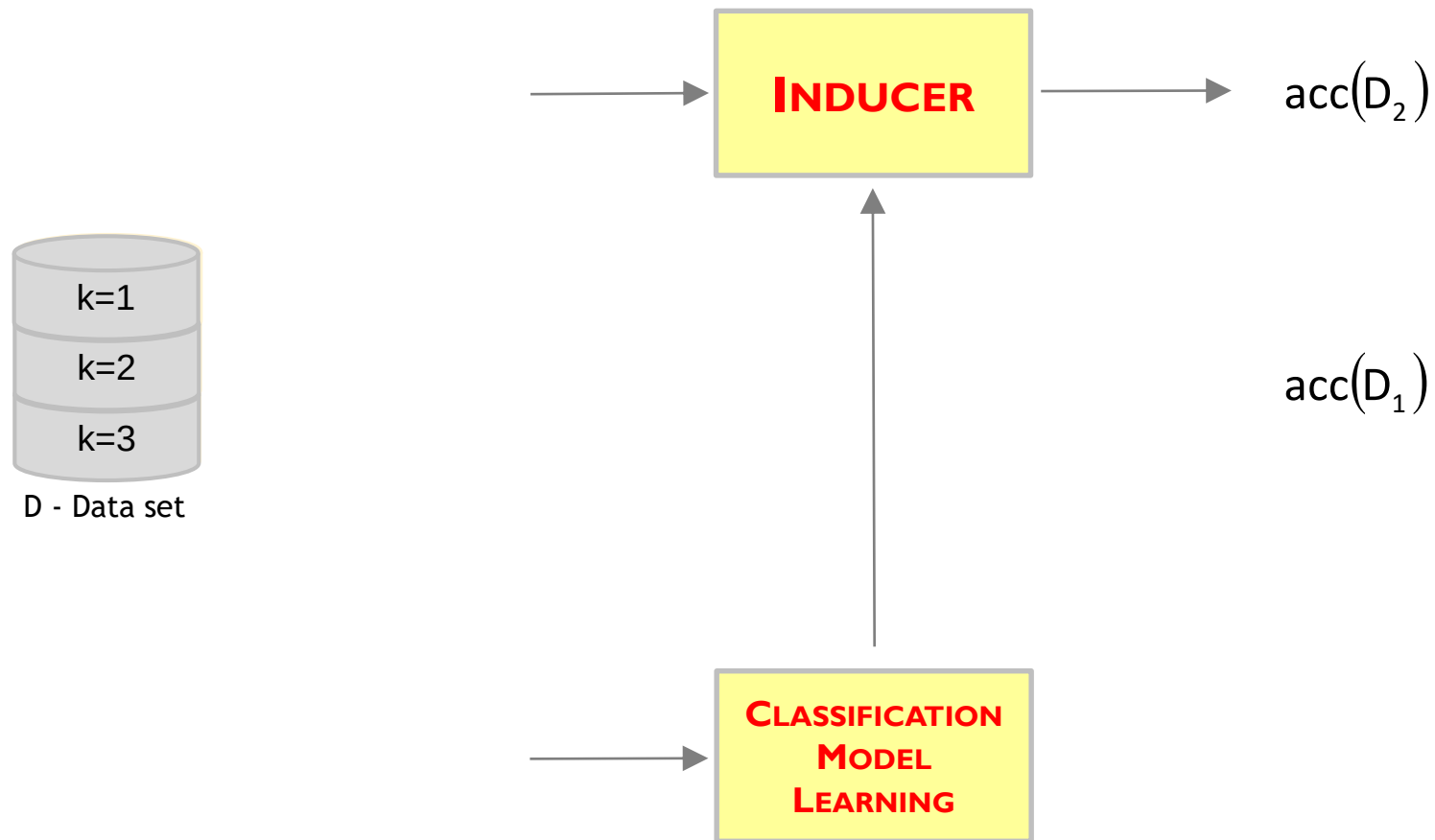
$$\text{acc} = \frac{1}{K} \sum_{k=1}^K \text{acc}(D_k)$$

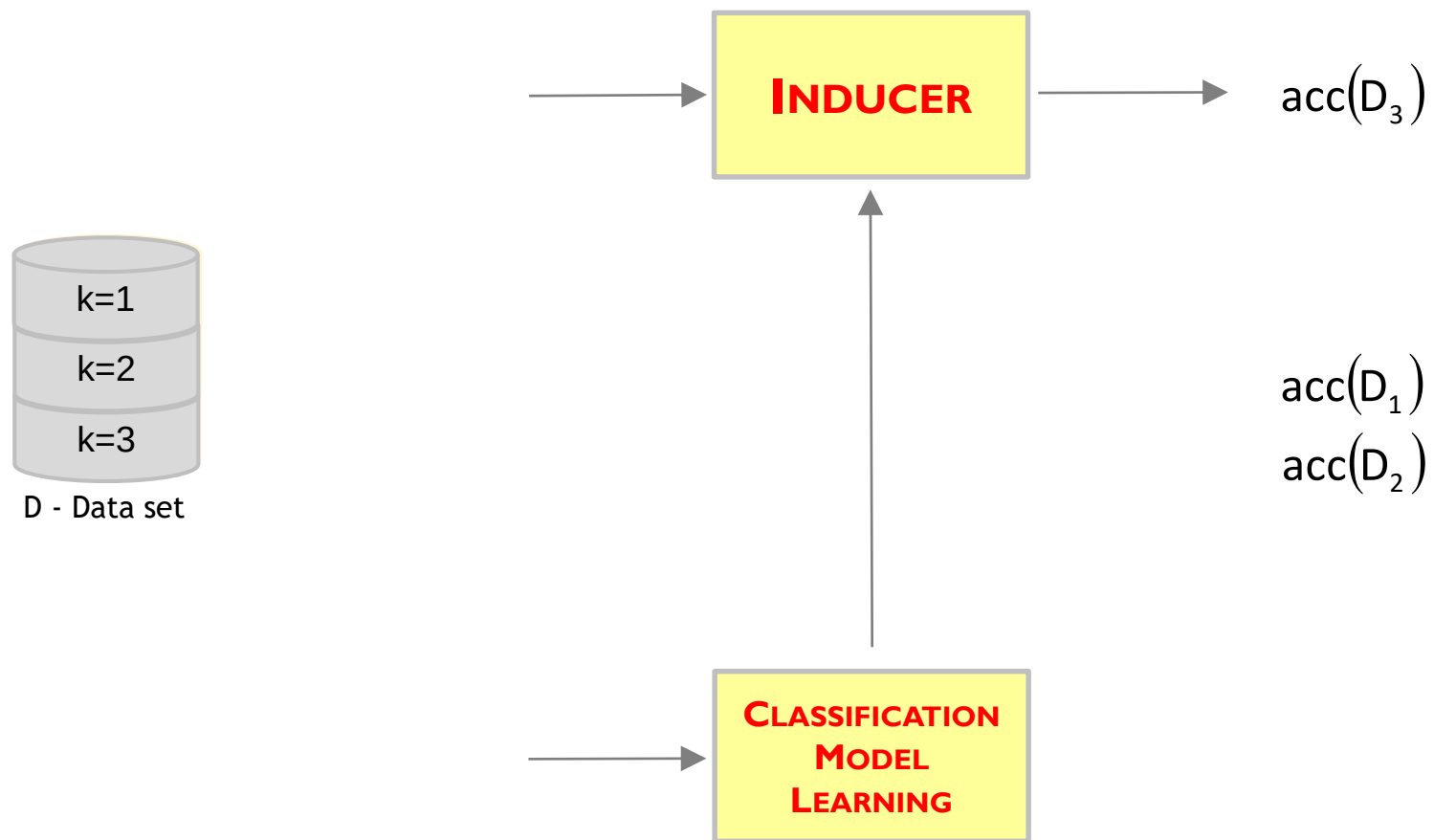
which offers a more effective estimates of the classifier's accuracy.

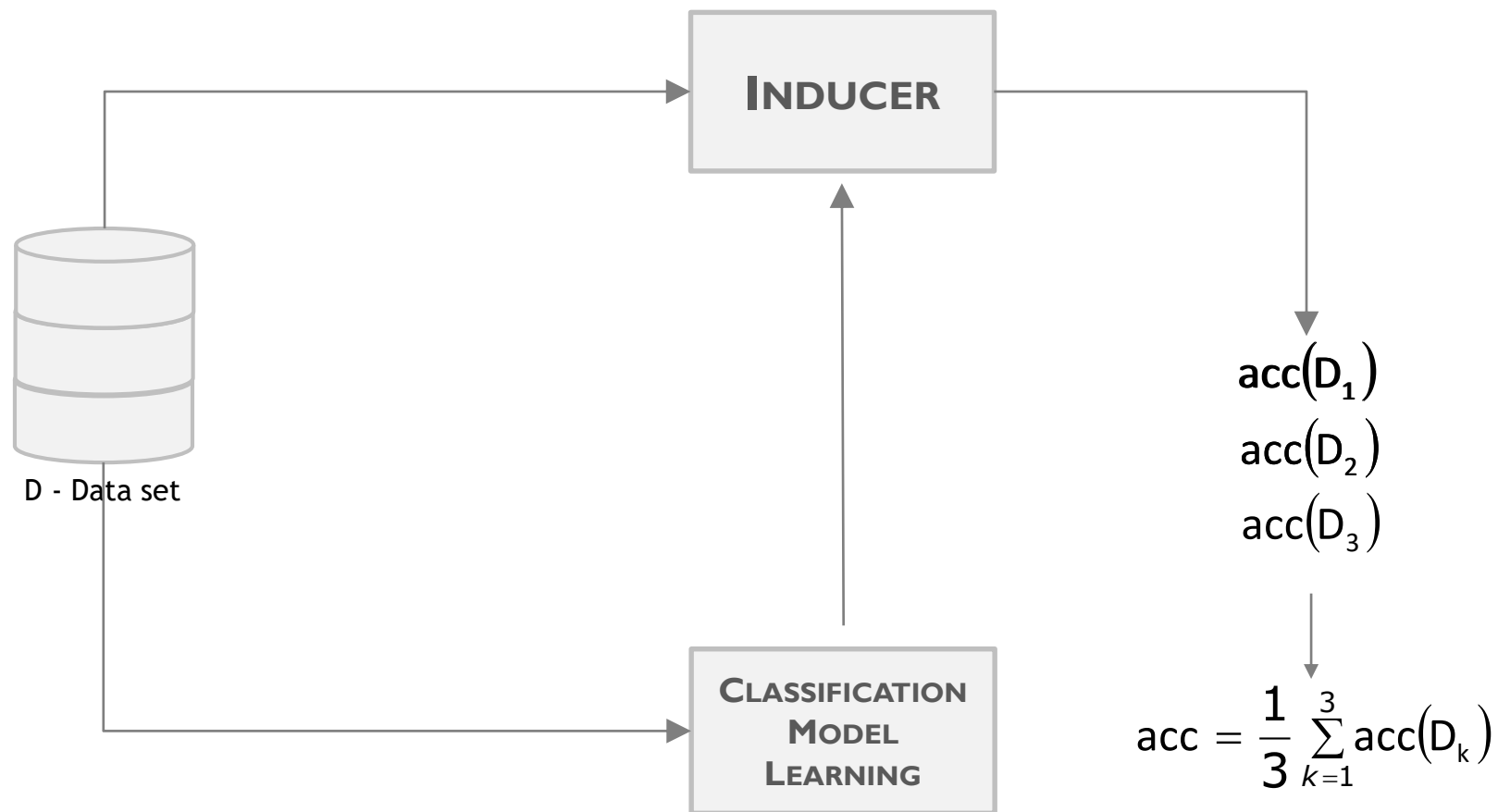
Different options exist to select the value of the parameter **K**. Typical values are **K = 3, 5, 10**, while a limit case applied when data are scarce is known under the name of **LEAVE ONE OUT CROSS VALIDATION (LOOCV)**.

LOOCV is obtained by assuming that each record is a partition of the dataset, thus the value of **K** equals the number of records of the dataset **D**.









We usually ask that **EACH PARTITION OF THE DATASET** contains the **SAME PROPORTION FOR POSSIBLE VALUES OF THE CLASS ATTRIBUTE**.

When class proportions are strongly different (unbalanced classes) we adopt specific sampling schemes, i.e. **STRATIFIED SAMPLING**.

STRATIFIED SAMPLING builds data subsets trying to ensure that classes proportion are maintained from train to test data sets. It can be proportional, equal probability, ...